Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/10/2005
PI Name:	Tittel, Frank K Ph.D.		
Project Title:	Development of Advanced Mid-Infrared Laser Based Gas Sensor Technology		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Special Category:	None		
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Zip Code:	77005	**Congressional District:**	25
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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COI Name (Institution):	Kosterev, Anatoliy A Ph.D. (Rice University) Bakhirkin, Yury Ph.D. (Rice University)		
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	This project is a joint Rice University- George Washington University-JPL research and development program of mid-infrared diode laser sources in the 3 to 4.4 micron region and their application to trace gas sensing in NASA applications relevant to spacecraft environmental monitoring and advanced life support. Presently, single frequency diode and quantum cascade lasers are not available in the proposed mid infrared spectral region that covers an important range where sensitive, selective and fast trace gas detection can now only be performed with complex spectroscopic sources, such as lead salt diode lasers and indirect laser sources based on difference frequency generation. Pulsed and continuous wave Type I quantum-cascade lasers, now commercially available cover the wavelength region from 4.4 - 24 microns. However, there is a significant need to access wavelengths from 3 to 4.4 microns, where a number of critical molecules (e.g. CH4, HCl, NO2, H2CO,SO2, and CO2) posses strong ro-vibrational lines. The Laser Science group at Rice University has accumulated a decade of experience in laser based trace gas detection in applications directly		

Task Description:	relevant to NASA as well as in other areas such as atmospheric chemistry, environmental monitoring, industrial process control and medical diagnostics (http://www.ece.rice.edu/). The group of Prof. Houston Miller is providing expertise in spectroscopic strategy for detecting Code M molecular target gases as well as developing a cavity ringdown spectroscopy (CRDS) based sensor technology. JPL with world-class semiconductor facilities and the expertise necessary is developing Sb-based diode and type II interband cascade lasers with distributed feedback structures for meeting the application needs. At Rice the primary emphasis will be to demonstrate and emphasize a novel optical detection technique based on an ultracompact photoacoustic analyzer design that will significantly enhance the performance and robustness of near infrared and mid-infrared gas sensors. The primary emphasis of this task is the development of compact, robust trace gas sensor technology based on quartz enhanced photoacoustic spectroscopy that can be applied to the sensitive, selective and real-time detection of Code M molecular targets at desired detection limits Manned space missions planned for this century will include longer stays on the International Space Station , expeditions to Mars, as well as return trips to the moon. . Hence there is a need to dreate and maintain an environment suitable for human habitation . This will require the development of instrumentation capable of precise detection, quantifying and controling a number of molecular trace gas species that are relevant to the air quality of spacecraft habitats.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Compact, sensitive and selective gas sensors based on near infrared and mid infrared semiconductor lasers and quartz enhanced photoacoustic spectroscopy. This technology is applicable to atmospheric Chemistry, environmental monitoring, chemical analysis and industrial prcess control as well as medical diagnostics based on breath analysis.
Task Progress:	The following tasks were addressed during calendar year 2004.<p> Formaldehyde detection using an interband cascade laser based on quartz enhanced photo-acoustic spectroscopy<p> The recent realization of DFB interband cascade lasers (ICLs) [1] has made it possible to access wavelengths between 3.0 and 4.5 µm, a spectral region which has been difficult to cover with intraband quantum cascade lasers (QCLs). The DFB-ICL used in this work emitted near 3.5 µm. This spectral region is important for various gas sensing applications because it corresponds to a spectral range of the C-H stretch vibration of aldehydes (e.g. formaldehyde symmetric stretch) and alkanes (e.g methane). Formaldehyde (H2CO) is of particular interest since it is a hazardous and carcinogenic substance, which is released from chemical binders present in numerous manufactured items and hence its presence in the environment cannot be avoided. The Occupational Safety and Health Administration (OSHA) has issued general industrial standards with an upper limit of 0.75 ppm for long term exposure (8 hours time-weighted average) and 2 ppm for short term exposure (15 minutes). NASA has also established spacecraft maximum allowable concentration levels for crew exposure to H2CO for extended periods of time. Furthermore, H2CO is an important and reactive component present in all regions of the atmosphere arising from the oxidation of biogenic and anthropogenic hydrocarbons. Tropospheric H2CO concentration measurements provide a means of validating photochemical model predictions concerning hydrocarbon oxidation that are used to advance ozone chemistry. To quantify H2CO concentrations, several different chemical and physical detection methods are used. Chemical analyzers, which employ coloration of a formaldehyde-sensitive reagent are sensitive at ppbv levels but they suffer from cross interference effects by other aldehydes and require long sampling times (i.e. minutes or more). To overcome these limitations, laser based spectroscopic sensors have been developed. The best H2CO detection sensitivity reported (< 50 pptv) was achieved using lead salt diode laser based absorption spectroscopy in a multipass optical cell with effective optical pathlength of 100 m. Currently, ICLs provide optical output powers of ~ 10 mW, a value which is expected to increase in the future. While the sensitivity of direct absorption spectroscopy does not fundamentally depend on laser power (neglecting the shot-noise limit), other approaches can make use of high spectroscopic source power to lower the detection limits. Photoacoustic spectroscopy (PAS) is one of such approach. PAS sensitivity scales linearly with the available laser power. A novel modification of PAS, called quartz-enhanced PAS (QEPAS) also permits to match the size of the laser source and the absorption detection module (ADM), both with a ~1cm characteristic linear dimension. In this work we report the development and performance characteristics of a formaldehyde sensor using a cw DFB-ICL and QEPAS. QEPAS is based on the photoacoustic signal build-up in a high-Q piezoelectrically active quartz crystal instead of a low-Q gas filled resonator as in traditional PAS. It has advantages of a very small sample volume (~ 1 mm3) required for analysis and high immunity to ambient acoustic noise. The sensitivity of QEPAS to a particular trace species is strongly dependent upon the V-T relaxation rate of this species in a certain host gas. If the rate is too slow, the thermal response giving rise to the detected pressure waves cannot follow the modulation frequency. A theoretical analysis of this rate is difficult and hence an experimental investigation is required to evaluate QEPAS sensitivity to specific chemical species. In this work we performed such studies for H2CO detection in dry nitrogen, dry nitrogen with 5% of SF6 and normal 50% humidity room air.<p> ICL parameters and H2CO absorption line selection: The ICL utilized in this work is able to operate continuous wave at temperatures of up to 170 K. The laser bias voltage is ~ 7 ? 8 V and the threshold current varies from 3 to 22 mA as the temperature changes from 78 K to 170 K. The laser frequency can be tuned continuously from 2833 cm-1 (T = 78 K) to 2816 cm-1 (T = 170 K) in a single frequency mode by means of current and temperature tuning. The temperature tuning rate is 0.178 cm-1 / K. At a fixed heatsink temperature of 78 K, the laser emits up to 12 mW and the output is tunable from 2831.8 cm-1 to 2833.7 cm-1 by current with a tuning rate of 0.0327 cm-1 / mA. The low threshold current operation of ICLs requires a current source with relatively high resolution (a ILX Lightwave Model LDX 3220 was used). The linewidth of a similar IC laser operating at 3.3 µm was measured to be < 20 MHz. With the available ICL operating at 78 K it is possible to access four significant formaldehyde absorption lines. For H2CO measurements in ambient air at concentrations <50 ppbv, a line at 2832.483 cm-1 was selected for this study and was accessed with an ICL drive current of 42 mA. <p> Description of the ICL based QEPAS Sensor Platform: The QEPAS ADM consisting of a crystal resonator/transducer and acoustic microresonator has been described in earlier publications [2-5]. Briefly, a watch tuning fork (TF) is used as the photoacoustic transducer. Two glass tubes (2.54 mm in length and 320 µm inner diameter) are arranged on both sides of the TF forming an acoustic microresonator, which enhances the QEPAS signal by a factor of ~10. The piezoelectric current generated by the excitation of the TF by photoacoustic wave action is converted to a voltage by a transimpedance amplifier (feedback resistor = 10 M?). ADM is placed into a gas cell with a volume of ? 1 cm3. Outer dimensions of the cell with electrical feedthroughs, pressure gauge, swagelock fittings and transimpedance amplifier mounted on the ISO-KF flange are 2.5in x 1.5 in x 2.5in. The ICL is mounted inside a liquid nitrogen (LN2) cooled cryostat (Cryo Industries, Inc). The ICL radiation (wavelength modulated at half the TF frequency of f ~ 32 kHz) is focused into ADM with a ZnSe lens (focal length 12.5 mm, 25 mm diameter). The QEPAS signal is subsequently demodulated by a lock-in amplifier (Stanford Research Model SR830 DSP) at frequency f (2nd harmonic detection) and processed by a laptop computer. A CaF2

beamsplitter directs 5% of the ICL beam through a short reference cell (5 cm long) filled with para-formaldehyde (which sublimes to H2CO with a vapor pressure of 1.45 Torr at 300 K). A photodetector signal is demodulated by a second lock-in amplifier at the 3rd harmonic of the f/2 reference and used as a feedback to lock the laser frequency to the H2CO absorption line via a proportional adjustment of the ICL current offset. A gas standard generator (Kin-Tek model 491M) based on a permeation tube was used to provide H2CO concentrations ranging from 0.5 to 25 ppmv in a diluting gas (i.e. N2 or air). The QEPAS cell pressure was maintained constant by means of a pressure controller (MKS Instruments Type 659), which also measured the gas flow (fixed at 75 sccm for all measurements with a needle valve located after the QEPAS cell). Since formaldehyde is a sticky molecule with a high dipole moment (2.3 D), ultra pure PFA (perfluoroalkoxy) tubing was employed in the gas flow system. This type of tubing has a smooth interior surface, thereby reducing adsorption of H2CO to the tubing walls. The QEPAS signal showed no dependence on the flow rate.<p> Experimental results:
 For good performance, it is essential to optimize two QEPAS parameters, namely the sampled gas pressure and the laser current modulation depth [2-5]. For this sensor the optimum gas pressure inside the QEPAS cell was found to be 200 Torr, with a TF resonance frequency of f = 32760 Hz and a Q factor of 16725. The optimum ICL current modulation amplitude at 200 Torr sample gas pressure was 4 mA corresponding to 0.13 cm-1 and roughly matching ~2 times the collision-broadened FWHM (0.06 cm-1 at 200 Torr) of the selected absorption line, similar to the previous QEPAS studies [4]. The addition of 5% SF6 to the N2 diluting gas enhanced the QEPAS signal by a factor of two. Ambient air was also found to result in a factor of 1.5 higher QEPAS signal when it was used as a diluter in place of dry N2. This effect is most likely due to its water content and the related increase of V-T energy transfer rate (the signal enhancement is not due to H2O absorption as this is one order of magnitude lower than that of H2CO).
 The H2CO concentration was set to 17.7 ppmv and ambient air was employed as the diluting gas. The ICL frequency was scanned in 20 MHz steps. The lock-in time constant was 1 s with a 3 s delay between two consecutive measurements. The minimum detection sensitivity of the sensor was evaluated from the 1 ? noise of the background and was determined to be 2.2 •10 8 cm 1 • W • (Hz) 1/2 for an IC laser power of 3.4 mW inside the cell (limited by the required spatial filtering of the applied laser beam and optical losses from the ZnSe lens and the sapphire windows of the QEPAS cell) and a peak formaldehyde absorbance of 6.6 • 10-5 cm-1. Stepwise concentration measurements were performed to verify the linearity of the QEPAS signal as a function of the H2CO concentration. The ICL was locked to the formaldehyde line at 2832.483 cm-1. The gas standard generator was used to produce H2CO concentrations in steps from 2 to 20 ppmv with ambient air as the diluting gas. The QEPAS signal for each concentration step was measured every 30 s for a total time duration of 10 minutes. The lock-in time constant was set to ?=10 s for these measurements.
 In summary, we have demonstrated the feasibility of using a novel cw 3.53 µm DFB ICL as a spectroscopic source for a QEPAS based sensor to detect formaldehyde concentrations at sub-ppmv levels. The measured detection sensitivity of the sensor is 2.2 •10 8 cm 1•W•(Hz) 1/2 This detection sensitivity corresponds to 0.55 ppmv H2CO concentration in air with 3.4 mW of laser power delivered to the QEPAS ADM and a time resolution of 10 s. It is expected that future ICL devices will be capable of emitting ~100 mW or more optical power. Assuming that such ICL performance is achieved and all the emitted power is delivered to the ADM, the QEPAS based H2CO detection sensitivity would be ~25 ppbv. Further gain in performance is anticipated with optimization of the micro-resonator design and by using lower frequency TFs. Thus, the combination of a powerful thermoelectrically cooled ICL and QEPAS ADM provide the technologies for compact field-deployable trace gas sensors. <p>

Instrument Design Team Meeting at NASA JPL.
 Three members of the Rice Laser Science Group participated in a 3 day Instrument Design meeting conducted by Robert J. Wilson and Greg H Bearman of JPL October 12-14, 2004.<p>

References:
 1.R.Q. Yang, Superlattices and Microstructures 17, 77 (1995).
 2.A.A. Kosterev, Y.A. Bakhirkin, F.K. Tittel, S. Blaser, Y. Bonetti, L. Hvozdara: Appl. Phys. B 78, 673 (2004).
 3.D. Weidmann, A.A. Kosterev, F.K. Tittel: Opt. Lett. 29, 1837 (2004).
 4.A.A. Kosterev, F.K. Tittel, Appl. Opt. 43, 6213-6217 (2004).

Bibliography Type:	Description: (Last Updated: 04/30/2007)
Articles in Peer-reviewed Journals	Kosterev, A.A.;Bakhirkin, Y.A; Tittel, F.K.; Blaser, S.; Bonetti, Y.; and Hvozdara, L. "Photoacoustic phase shift as a chemically selective spectroscopic parameter" N/A , Jul-2004
Articles in Peer-reviewed Journals	Weidmann, D.; Kosterev, A. A.; Roller,C.; Curl, R.F.; Fraser M.P.; Tittel, F. K. "Monitoring of ethylene by a pulsed Quantum Cascade Laser" N/A , Jul-2004
Articles in Peer-reviewed Journals	Weidmann, D.; Kosterev, A.A.; Tittel,F.K.; Ryan, N.; McDonald, D. "Application of widely electrically tunable diode laser to chemical gas sensing with quartz-enhanced photoacoustic spectroscopy" N/A , Jul-2004
Articles in Peer-reviewed Journals	Kosterev, A.A.; Tittel, F.K. "Ammonia detection by use of quartz-enhanced photoacoustic spectroscopy with a near-IR telecommunication diode laser" N/A , Jul-2004
Articles in Peer-reviewed Journals	Miller, J.H.; Awtry, A.R.; McAndrew, B.; Tittel , F.K.; Kosterev, A.A.; Yang, R. Q.; Hill, C.J.; Chung M. Wong, Baohua Yang, and Greg Bearman "Development of Optical Trace Gas Monitoring Technology for NASA Human Space Flight" N/A , Jul-2004
Articles in Peer-reviewed Journals	Weidmann, D.; Tittel, F.K.; Aellen, T.; Beck, M.; Hofstetter, D.; Faist, J.; Blaser, S. "Mid infrared trace gas sensing with a quasi-continuous wave Peltier-cooled distributive feedback quantum cascade laser" N/A , Nov-2004
Articles in Peer-reviewed Journals	Horstjann, M.; Bakhirkin, Y.A.; Kosterev, A.A.; Curl, R.F.; Tittel, F.K. "Formaldehyde sensor using interband cascade laser based quartz-enhanced photoacoustic spectroscopy" N/A , Nov-2004
Presentation	Kosterev, A.A.; Bakhirkin, Y.; Tittel, F.K. "Advanced detection techniques of trace gas monitoring for human space flight" Habitation 2004 International Conference, Orlando, FL (January 4-7, 2004). Jan-2004
Presentation	Kosterev, A.A; Tittel, F.K. "Recent advances in quartz-enhanced gas-phase photoacoustic spectroscopy" OSA Topical Meeting on Laser Applications to Chemical and Environmental Analysis, (LACEA), Annapolis, MD, (February 9-11, 2004). Feb-2004

Presentation	Kosterev, A.A.; Weidmann, D.D; Tittel, F.K.; Ryan, N.; McDonald, D. "Multi-species QEPAS based gas sensing using a wavelength-programmable diode laser source" Conference on Lasers and Electro-Optics/International Quantum Electronics Conference (CLEO/IQEC 2004), San Francisco, CA (May 16-21, 2004). May-2004
Presentation	Miller, J. H.; Awtry, A. R.; McAndrew, B.; Tittel, F. K.; Kosterev, A. A.; Yang, R. Q.; Hill C. J., Wong, C. M.; Yang, B.; Bearman, G. "Development of optical trace gas monitoring technology for NASA human space flight" 34th International Conference on Environmental Systems (ICES 2004), Colorado Springs, CO, (July 19-22, 2004). Jul-2004
Presentation	Kosterev, A.A.; Bakhirkin, Y. A.; Tittel, F. K. "Quartz-enhanced photoacoustic spectroscopy with semiconductor lasers: the road to ultracompact trace gas sensor" Novosibirsk, Russia, 4th International Symposium on Modern Problems of Laser Physics, (MPLP 2004), Novosibirsk, Russia, (Aug. 22-27, 2004) Aug-2004
Presentation	Tittel, F.K.; Bakhirkin,Y; Curl, R.F.; Kosterev, A.A.; So, S.; Weidmann, D.; Wysocki, G. "Development of quantum cascade laser based sensor technology for trace gas monitoring applications" 5th Quantum Cascade Laser Workshop, Fraunhofer Institute for Physical Measurement Techniques, Freiburg, Germany, (Sept. 23-24, 2004) Sep-2004
Presentation	Tittel, F. K.; Kosterev, A.A.; Bakhirkin, Y. "Recent advances in quartz-enhanced gas-phase photoacoustic spectroscopy" 3rd IEEE Intl. Conference on Sensors, Vienna, Austria (Oct. 24-27, 2004). Oct-2004